

# Irregular Sampling Analysis: Layout Effects on Estimation

STAT778 Project

2026-04-24

## Contents

|                          |   |
|--------------------------|---|
| Purpose . . . . .        | 1 |
| Load and tidy . . . . .  | 1 |
| MAE by layout . . . . .  | 3 |
| Bias by layout . . . . . | 4 |
| Summary . . . . .        | 5 |

## Purpose

Compares estimation accuracy under three spatial layouts:

- **Grid:** regular 7x7 (K13-A default).
- **Random:** 49 uniform points in  $[0,1]^2$ .
- **Clustered:** 49 points in 4 Gaussian clusters.

All share the same generating parameters (Poisson  $\lambda=5$ ,  $\text{range}=0.3$ ). Any difference in estimation accuracy therefore reflects the layout itself.

---

## Load and tidy

```
source("code/R/logging.R")
source("code/R/io.R")
source("code/R/fit_core.R")

DATASET <- "IRREGULAR"
ids <- list_scenarios(DATASET, dir = "data/generated")
fits <- lapply(ids, function(id) load_fit_result(DATASET, id))
names(fits) <- ids
length(fits)
```

```
## [1] 3
```

## Visualize the three layouts

Before analysis, confirm the layouts look as intended.

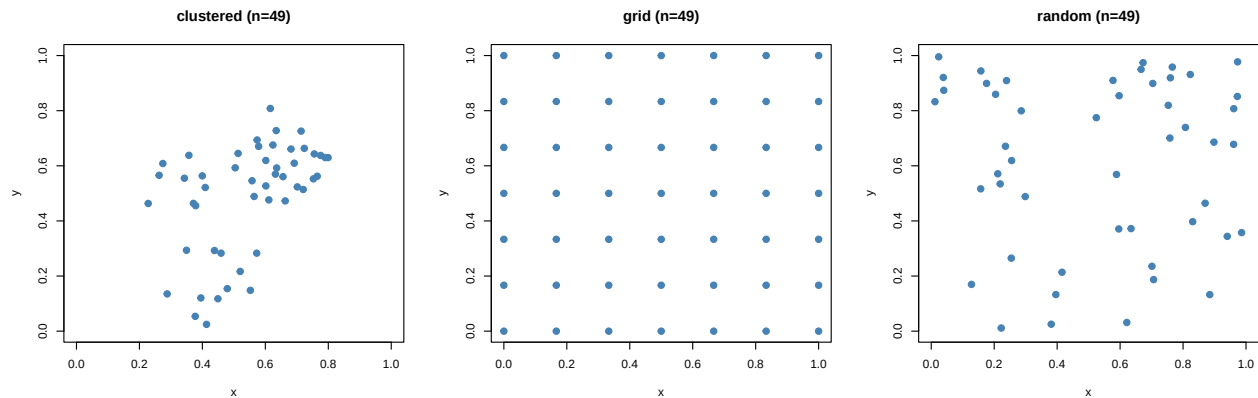
```
par(mfrow = c(1, length(fits)))

for (id in names(fits)) {
  locs <- fits[[id]]$scenario_config$layout
  scen <- load_scenario(DATASET, id, dir = "data/generated")
}
```

```

plot(
  scen$locations[, "x"], scen$locations[, "y"],
  pch = 19, cex = 1.2, col = "steelblue",
  xlab = "x", ylab = "y",
  xlim = c(0, 1), ylim = c(0, 1),
  main = sprintf("%s (n=%d)", id, nrow(scen$locations))
)
}

```



```

par(mfrow = c(1, 1))

```

## Tidy

```

tidy_one <- function(f) {
  r <- f$results
  cfg <- f$scenario_config

  rows <- list()
  for (i in seq_len(nrow(r))) {
    if (!r$converged[i]) next
    est <- r$estimates[[i]]
    if (is.null(est) || length(est) == 0) next
    for (pname in names(est)) {
      rows[[length(rows) + 1]] <- data.frame(
        scenario_id = f$scenario_id,
        layout_type = cfg$layout$layout_type,
        replicate   = r$replicate[i],
        method      = r$method[i],
        runtime_sec  = r$runtime_sec[i],
        parameter    = pname,
        estimate     = as.numeric(est[[pname]]),
        stringsAsFactors = FALSE
      )
    }
  }
  do.call(rbind, rows)
}

long <- do.call(rbind, lapply(fits, tidy_one))

```

```

long$truth <- NA_real_
long$truth[long$parameter %in% "Intercept"] <- log(5)
long$truth[long$parameter %in% "range"] <- 0.3

long$abs_err <- abs(long$estimate - long$truth)
long$err <- long$estimate - long$truth

```

## MAE by layout

```

mae_tab <- aggregate(
  abs_err ~ scenario_id + method + parameter,
  data = long,
  FUN = mean
)
names(mae_tab)[4] <- "MAE"
mae_tab[order(mae_tab$parameter, mae_tab$scenario_id, mae_tab$method), ]

```

```

##   scenario_id method parameter      MAE
## 1   clustered   GHK Intercept 0.20494072
## 4   clustered   GQT Intercept 0.19976582
## 2     grid     GHK Intercept 0.14221800
## 5     grid     GQT Intercept 0.13992661
## 3   random     GHK Intercept 0.15101476
## 6   random     GQT Intercept 0.15026109
## 7   clustered   GHK   range 0.06052719
## 10  clustered   GQT   range 0.05810206
## 8     grid     GHK   range 0.06391373
## 11    grid     GQT   range 0.06265425
## 9   random     GHK   range 0.06381540
## 12   random     GQT   range 0.06101742

```

## Figure

```

par(mfrow = c(1, 2))

for (param in c("Intercept", "range")) {
  d <- mae_tab[mae_tab$parameter == param, ]
  layouts <- c("grid", "random", "clustered")
  d$scenario_id <- factor(d$scenario_id, levels = layouts)

  wide <- reshape(d[, c("scenario_id", "method", "MAE")],
    idvar = "scenario_id", timevar = "method",
    direction = "wide")

  mat <- rbind(wide$MAE.GHK, wide$MAE.GQT)
  colnames(mat) <- wide$scenario_id
  rownames(mat) <- c("GHK", "GQT")

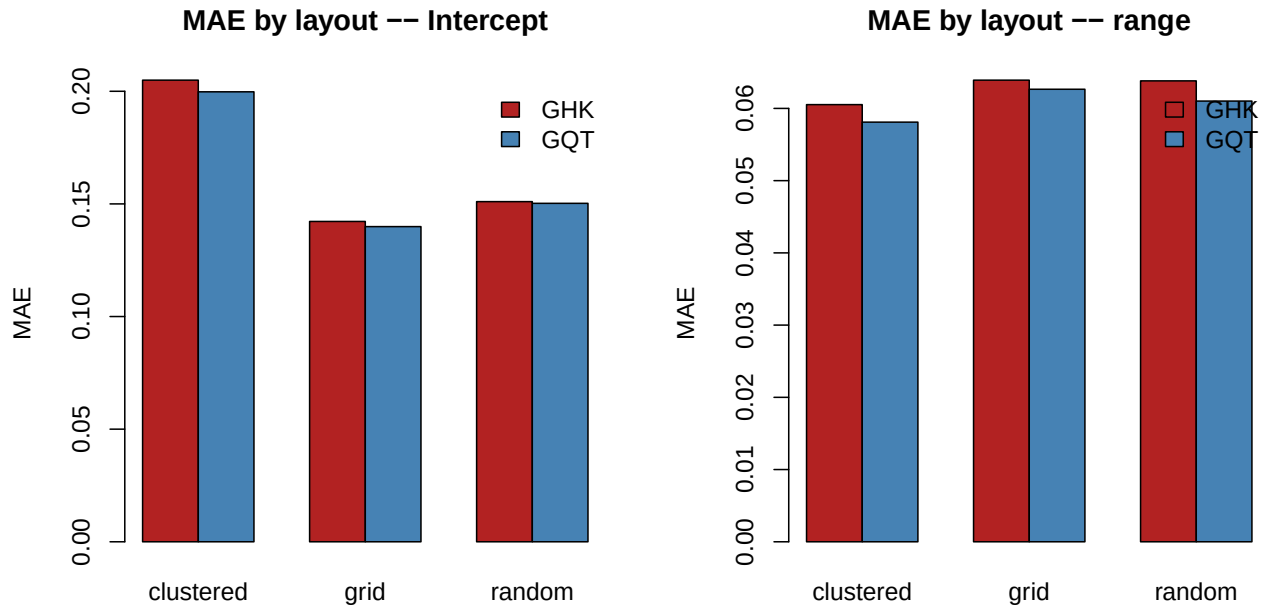
  barplot(
    mat, beside = TRUE,
    col = c("firebrick", "steelblue"),

```

```

ylab = "MAE", main = sprintf("MAE by layout -- %s", param),
legend.text = rownames(mat),
args.legend = list(x = "topright", bty = "n")
)
}

```



```

par(mfrow = c(1, 1))

```

## Bias by layout

```

bias_tab <- aggregate(
  err ~ scenario_id + method + parameter,
  data = long,
  FUN = mean
)
names(bias_tab)[4] <- "bias"
bias_tab

```

| ##    | scenario_id | method | parameter | bias          |
|-------|-------------|--------|-----------|---------------|
| ## 1  | clustered   | GHK    | Intercept | -0.0218651079 |
| ## 2  | grid        | GHK    | Intercept | -0.0001080704 |
| ## 3  | random      | GHK    | Intercept | 0.0014210703  |
| ## 4  | clustered   | GQT    | Intercept | -0.0087699191 |
| ## 5  | grid        | GQT    | Intercept | 0.0011214702  |
| ## 6  | random      | GQT    | Intercept | 0.0030711828  |
| ## 7  | clustered   | GHK    | range     | -0.0071171397 |
| ## 8  | grid        | GHK    | range     | 0.0052004738  |
| ## 9  | random      | GHK    | range     | 0.0053281417  |
| ## 10 | clustered   | GQT    | range     | -0.0213461660 |
| ## 11 | grid        | GQT    | range     | 0.0094860137  |
| ## 12 | random      | GQT    | range     | 0.0031702568  |

Regular grid baseline: bias should be near zero for both parameters. Random layout: expected similar.

Clustered: might show more bias (especially in range) because clustered point patterns have a different distribution of pairwise distances.

---

## Summary

```
cat("=== Irregular sampling summary ===\n\n")

## === Irregular sampling summary ===
for (layout in c("grid", "random", "clustered")) {
  cat(sprintf("\n-- %s --\n", layout))
  for (param in c("Intercept", "range")) {
    for (m in c("GHK", "GQT")) {
      mae_val <- mean(long$abs_err[long$scenario_id == layout &
                                long$parameter == param &
                                long$method == m], na.rm = TRUE)
      cat(sprintf("  %-9s  %s  MAE=%.4f\n", param, m, mae_val))
    }
  }
}

##
## -- grid --
##   Intercept  GHK  MAE=0.1422
##   Intercept  GQT  MAE=0.1399
##   range      GHK  MAE=0.0639
##   range      GQT  MAE=0.0627
##
## -- random --
##   Intercept  GHK  MAE=0.1510
##   Intercept  GQT  MAE=0.1503
##   range      GHK  MAE=0.0638
##   range      GQT  MAE=0.0610
##
## -- clustered --
##   Intercept  GHK  MAE=0.2049
##   Intercept  GQT  MAE=0.1998
##   range      GHK  MAE=0.0605
##   range      GQT  MAE=0.0581
```

## Likely findings

Fill in:

1. All three layouts maintain \_\_\_\_% convergence.
2. Random layout produces MAE \_\_\_\_ (similar to / worse than) grid.
3. Clustered layout \_\_\_\_ (the expected issue is worse range identification, since clustered points have fewer large distances to probe the correlation at its tail).